Heat is not temperature

antocciphysics | TEACHER KEY | CPA

Learning target: Distinguish temperature from energy transferred because of a temperature difference.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 20; 3 course days later.*
No; it lacks 3 J of the required energy and remains bound under the model.

02. Explain a worked example (worked)
Same-temperature samples have no net heat flow due to a temperature difference; larger amount generally stores more internal energy for the same material/state.

03. Look and explain (model)
Temperatures equal; larger amount generally has more internal energy under comparable conditions.

04. Build the solution (practice)
From block to water; block cools, water warms.

05. Use the idea (reason)
Thermal energy transfers from hotter drink to cooler surroundings.

06. Choose a method (mixed) *Review: Unit 4, Day 11; 12 course days later.*
4 J of rotational kinetic energy. A rolling wheel also spins.

07. Exit check (exit)
Temperature in°C or K; heat transfer in J. Temperature is not an amount of heat.

08. STRETCH: Apply it another way (stretch)
No; need mass, material and state/reference for an energy comparison.

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01. Bring back an earlier idea (retrieve) *Review: Unit 4, Day 20; 3 course days later.*
New speed is 1/3; $v_{escape}=\sqrt{(2\ GM/r)}$.

02. Explain a worked example (worked)
Same-temperature samples have no net heat flow due to a temperature difference; larger amount generally stores more internal energy for the same material/state.

03. Represent and interpret (model)
Temperatures equal; larger amount generally has more internal energy under comparable conditions.

04. Solve and justify (practice)
Three times as much energy for the larger sample because $Q=mc\Delta T$.

05. Analyze the reasoning (reason)
Thermal energy transfers from hotter drink to cooler surroundings.

06. Choose a method (mixed) *Review: Unit 4, Day 11; 12 course days later.*
$v=\sqrt{(2\ gh)}\approx 4$ m/s.

07. Exit check (exit)
Different amounts of material and different material/phase or microscopic energy structure.

08. STRETCH: Apply it another way (stretch)
40°C; equal heat capacities per sample make gains/losses symmetric. Unequal masses need an energy balance.